계산 물질 물리 연구실



김봉재교수

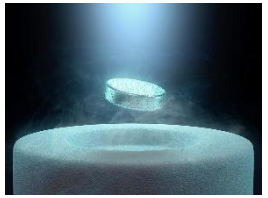
사무실: 326호

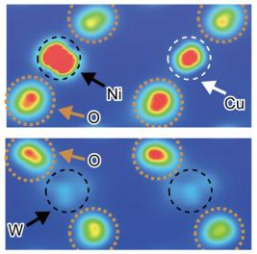
이메일: bongjae@knu.ac.kr

오피스: 053-950-5324

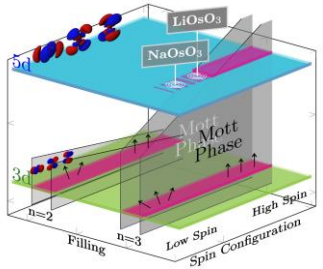
다양한 물질의 성질의 이해

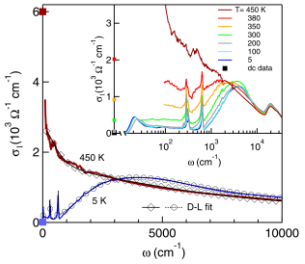
superconductivity



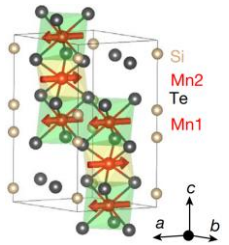


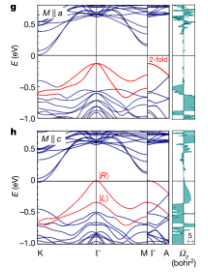
correlated systems

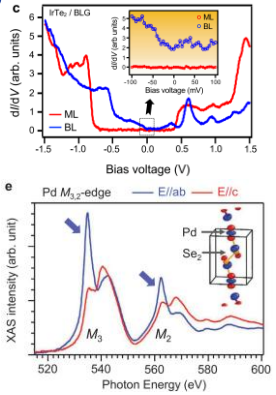




2D systems, ...

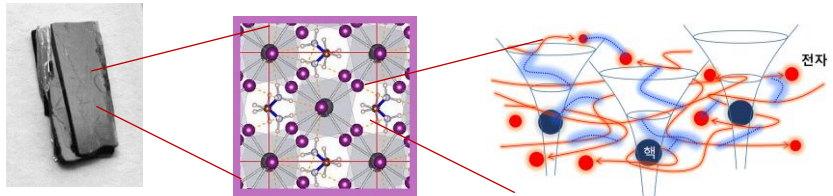




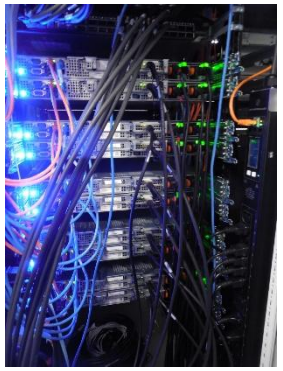


- **목표1**: 다양한 물질의 성질을 이해하는 것 - 초전도, 자성, 저차원, 광학 물질, 배터리 등

물질 내 전자의 운동 방정식

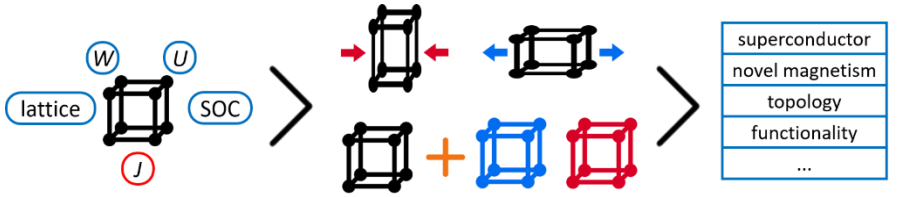


$$\hat{H} = -\frac{\hbar^2}{2m_e}\sum_{i=1}^{N}\nabla_i^2 - \sum_{n=1}^{M}\frac{\hbar^2}{2M_n}\nabla_n^2 + \frac{1}{4\pi\epsilon_0}\frac{1}{2}\sum_{i,j=1;i\neq j}^{N}\frac{e^2}{|\mathbf{r}_i-\mathbf{r}_j|}$$
$$-\frac{1}{4\pi\epsilon_0}\sum_{n}^{M}\sum_{i}^{N}\frac{Z_n e^2}{|\mathbf{r}_i-\mathbf{R}_n|} + \frac{1}{4\pi\epsilon_0}\frac{1}{2}\sum_{n,m=1;n\neq m}^{M}\frac{Z_n Z_m e^2}{|\mathbf{R}_n-\mathbf{R}_m|}$$

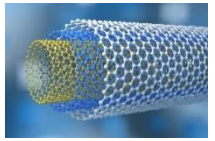


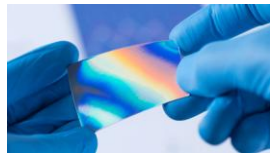
- **어떻게?** 전자의 운동방정식을 푼다. 컴퓨터를 이용하여.

물질 디자인, 물성 예측





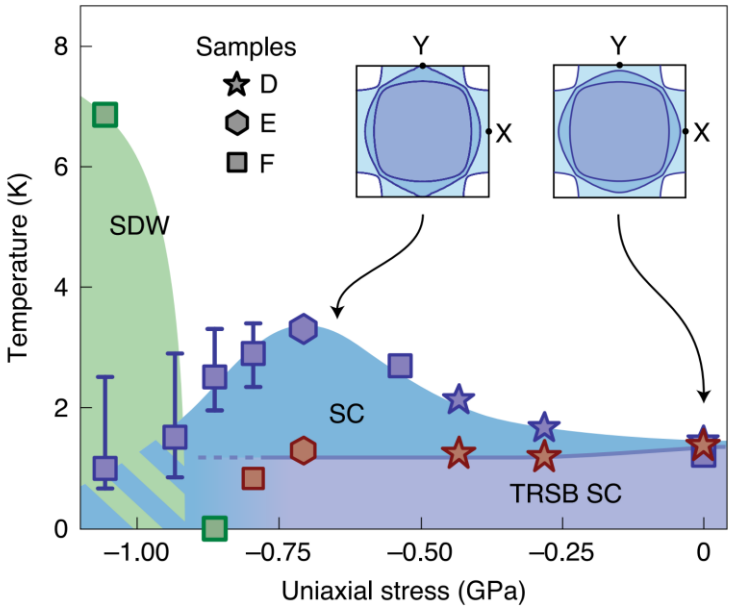




- **목표2**: 신물질을 디자인, 예측하는 것

물성의 이해: 왜 그런가?

superconductor



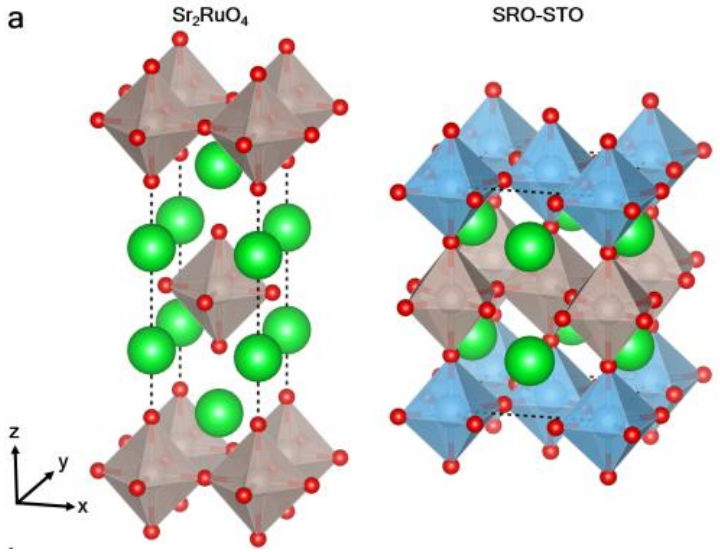
☐ 초전도의 원인은 무엇인가?

☐ 초전도의 특징은 무엇인가?

☐ 왜 (특정한) 실험 결과가 나오는가?

☐ 시뮬레이션을 통해 컴퓨터 실험 수행 후 대답.

☐ 시뮬레이션을 통해 컴퓨터 실험 수행 후 예측.

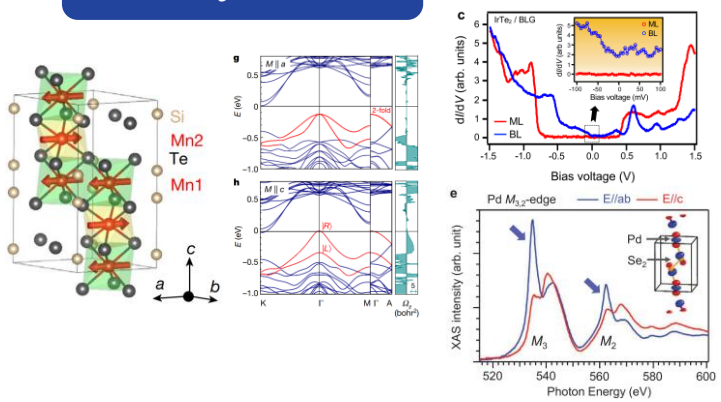


☐ 초전도를 만들어 낼 수 있을까?

☐ 기존 물질의 어떤 특성이 중요할까?

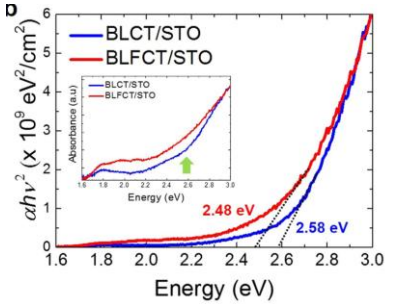
물성의 이해: 왜 그런가?

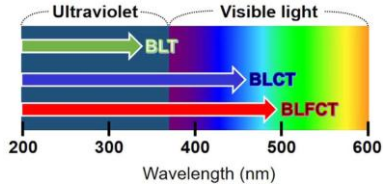
2D systems



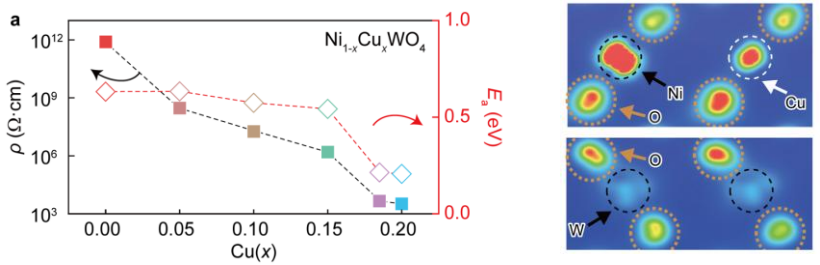
✓ 2D magnetism, topology, charge-order

Photovoltaic



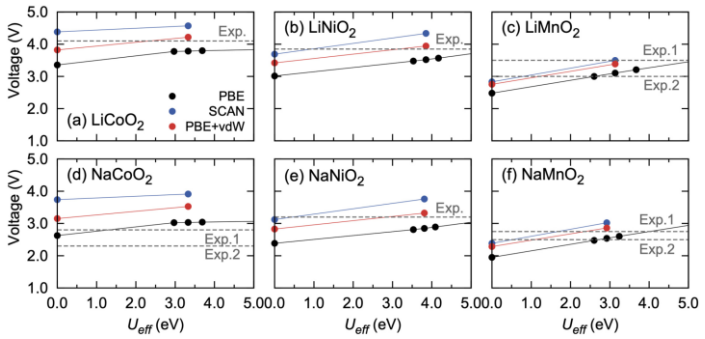


Optoelectronic device



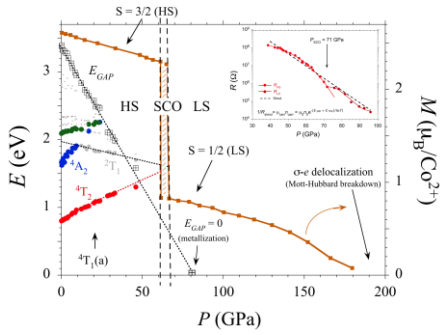
✓ opto-device, p-type semiconductor

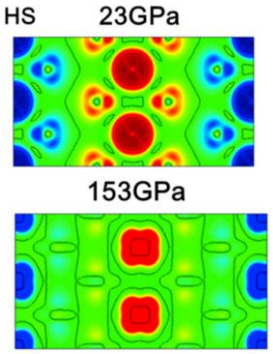
Battery



✓ Li/Na-ion battery

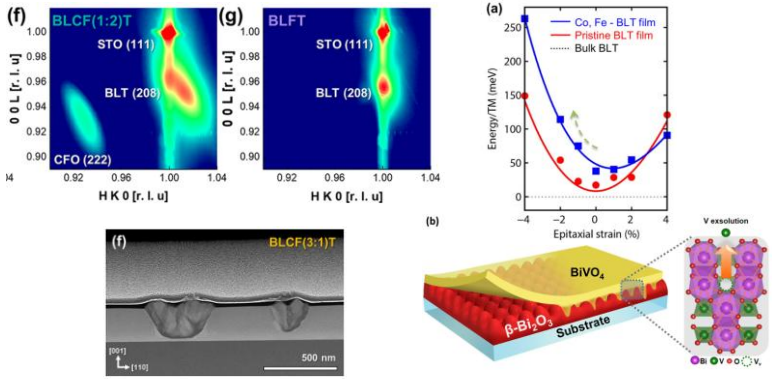
High pressure





✓ structural, magnetic, electronic phase transition

Growth behaviors



✓ exsolution, phase separation